Quantifying and Forecasting the Impact of the 2025 U.S. Government Shutdown on New York–Metro Flight Delays

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Objective

Measure how the 2025 shutdown affected New York–Metro air travel:

1. **Quantify** the change in delays and cancellations
2. **Test** whether weather explains the change
3. **Forecast** delays from information known in advance

Background

43 shutdown days | **40** airports cut | **3** airports studied

WHAT HAPPENED

- Longest U.S. shutdown on record
- Air-traffic controllers worked **without pay**; staffing shortages grew

Nov 7 — FAA cuts flights [3]

Oct 1 — Shutdown begins

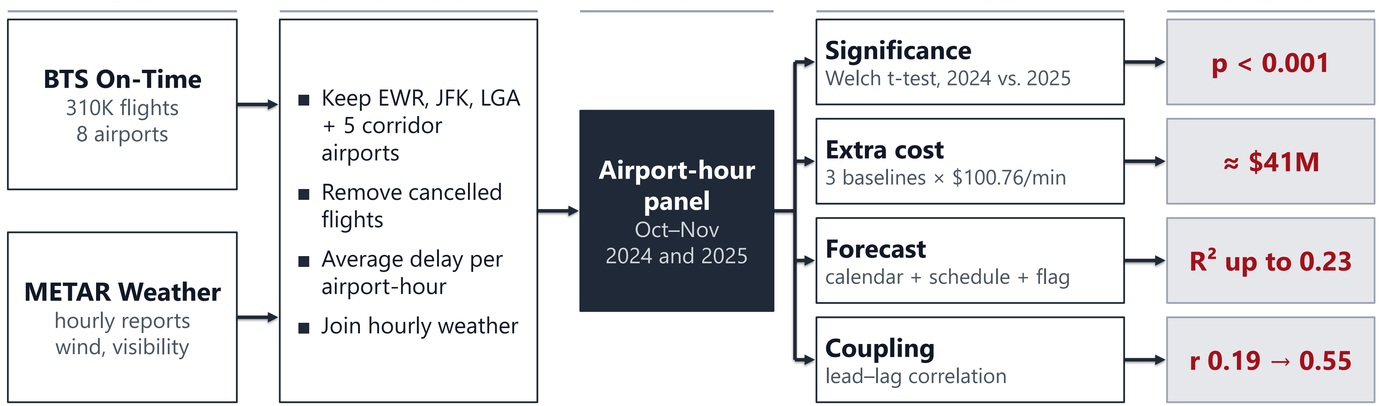
Nov 12 — Shutdown ends

WHY IT MATTERS

- NY has the most congested U.S. airspace
- Delay costs airlines **$100.76 per minute** [1]
- Delays spread between airports [2]

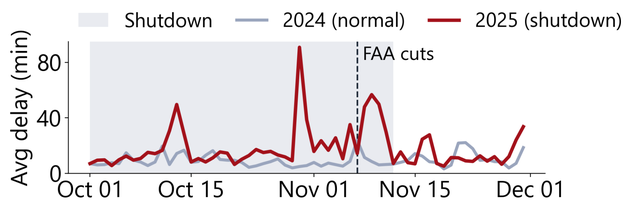
Research gap: no data-driven estimate of the shutdown's effect on New York–Metro flights

Methodology



Result 1: Delays Over Time

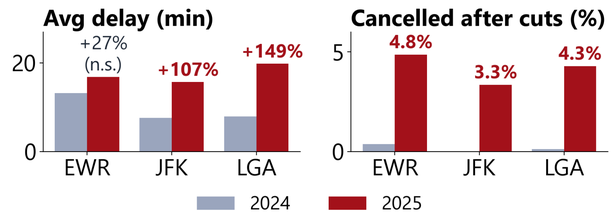
Delays spiked during the shutdown



Daily average departure delay, EWR + JFK + LGA

Result 2: By Airport

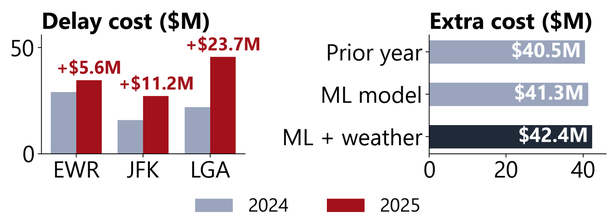
JFK and LGA delays more than doubled



JFK, LGA: $p < 0.001$ · EWR: $p = 0.07$ (n.s.)

Result 3: Extra Delay Cost

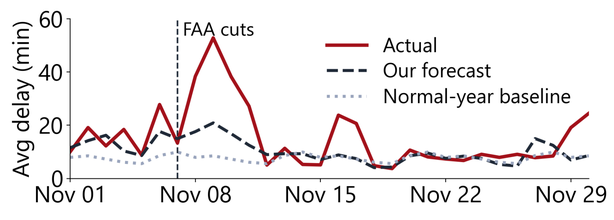
≈ $41M extra, not explained by weather



Cost at $100.76 per delay minute [1]

Result 4: Forecast

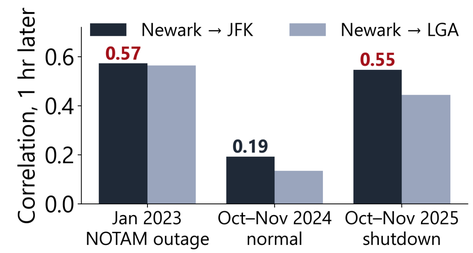
Our forecast beat the normal-year baseline



R^2, forecast vs. baseline: EWR .23/.09 · JFK .12/−.01 · LGA .15/−.04

Result 5: Coupling

Under stress, Newark's delays moved with JFK and LGA



1-hour lagged correlation (does not imply causation)

Conclusions

- **JFK +107%, LGA +149%**; EWR +27% not significant
- **≈ $41M** extra cost, not from weather
- Partly **predictable** from public data
- Airports **more linked** under stress

Limits: departures only; one prior year

Next: targeted vs. blanket flight cuts

References

[1] Airlines for America, Delay Costs, 2025.
[2] Fleurquin et al., Sci. Reports 3:1159, 2013.
[3] FAA Emergency Order, Nov. 6, 2025.
[4] BTS, On-Time Performance data.
[5] Iowa Environmental Mesonet, ASOS/METAR.

Research Repository
Code, data & figures
github.com/pranishuprety/stemadvance2026